

Study design

Omics data

Data analysis

N = 57

N = 154

Liver
Biopsy

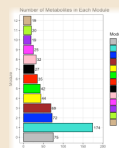
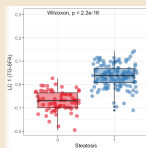
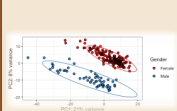
Serum



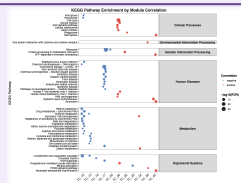
Liver biopsy

Serum

Transcriptomics Lipidomics Metabolomics Metabolomics

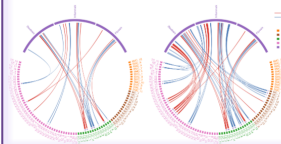


Integration



Men

Women



Key findings:

Liver Transcriptomics

- Amino acid metabolism ↓

Liver Metabolomics

- Amino acids ↓, fatty acids remodeled

Serum metabolites

- Amino acids ↑ (systemic spillover from impaired hepatic metabolism)

Liver Lipidomics

- Triacylglycerols (TAG) ↑, phospholipids (PC, PE) ↓ during MASLD progression

Cross-omics

- Lipid modules mediate transcriptomic effects on liver pathology

Sex Differences

- Men show a focused TAG pathway
- Women show broader multi-lipid regulation and cross-talk